Gray-Zone Maritime Control in the First Island Chain: A Networked Escalation Game

Updates to the Workshop 1 Writeup (working copy for Workshop 2)

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This is a copy of the Workshop 1 report (as submitted, branch `workshop1-final`) with the Workshop 2 model changes written in. Text in this color is new or changed. ~~Struck-out red text~~ is removed. Sections not yet touched are unchanged from Workshop 1, including all results, which will be replaced after the rerun.

1 Setting and research questions

Contested system: China and the United States compete, *below the threshold of war*, for day-to-day control of the waters inside the First Island Chain. China claims most of the South China Sea and contests the Senkakus, pressing these claims through the persistent presence of coast guard and maritime militia (CCG) hulls backed by navy (PLAN) ships. The U.S. rejects maritime claims beyond what international law allows and keeps the seas open with a smaller number of destroyers (DDG) on presence and freedom-of-navigation patrols (Helmus et al., 2024; Green et al., 2017). Every two days, each side decides *where* to station its ships on a network of zones and *how* to act there: patrol (P) or assert (A: shadowing, close maneuvering, querying and warning, and, for the U.S. only, freedom-of-navigation operations).

Why strategic interaction is central: control of a zone depends on the ratio of local forces, so each side's best allocation depends on the other's. Escalation risk is shared. Incidents need both sides present, and war costs both. Neither side sees the other's entire fleet disposition, only forces present where they also maintain forces.

Research questions:

RQ1. *Control vs. war risk.* Each side caps how much incident risk it will accept each period (its risk budget ε_i). How much more control does a side gain by accepting more risk, how much more likely does war become, and what risk level does each side pick when it knows the other side is picking too?

RQ2. *Game theory vs. rules of thumb.* Does a side that plans with game theory (best responses, equilibrium or maximin plans) do better than one that follows simple rules or moves at random? Can both sides keep patrolling over many periods because asserting once would end the truce and push both into riskier play from then on?

RQ3. *Weather and redesign.* Does the plan or Blue redesign that works best in normal weather still work best across calm, typhoon and extreme weather scenarios, both on average and in the worst scenario?

Outcomes (averaged over 200 simulated games per setup): Blue's value-weighted control share $\bar{c}_{\mathsf{B}} = \frac{1}{\tau}\sum_t \sum_v V_v^{\mathsf{B}} c_v^t / \sum_v V_v^{\mathsf{B}} \in [-1,1]$, war rate and payoffs $u_{\mathsf{B}}, u_{\mathsf{R}}$. For RQ3, also expected and worst-case payoff over ω.

Changes from the proposal and how we used the feedback: the model is now a formal game on a graph (§3). The best response is an exact MIP whose feasible region is a risk budget, with the proposal's commander types as budget levels (§3.7). The RQs now ask for the budget-game equilibrium, equilibrium play, and robustness and redesign. We define three Blue redesigns (§3.10), propose values for missing data (Tables 1–2), report analytical results and 77 configurations × 200 seeds (§4), and add weather scenarios scored by expected and worst-case payoff, as Prof. Sapsis asked (§3.2, §4.4).

2 Literature and current practice

Documented practice: Helmus et al. (2024) describe China's use of coast guard and maritime militia vessels to maintain presence and obstruct maritime activities while seeking to avoid a conventional military response. Green et al. (2017) emphasize credible commitments, coordinated allied responses, and acceptance of calculated escalation risk while exercising restraint. U.S. practice is routine presence and freedom-of-navigation patrols, with encounters managed by the Code for Unplanned Encounters at Sea (WPNS, 2014). From these sources we take the actions, the asymmetric hull types, and the idea that incidents, not shots, drive escalation.

Models: Wasser et al. (2019) developed a structured gray-zone game with explicit objectives and probabilistic adjudication. We automate the rules to support repeated simulation and strategy comparison. Shieh et al. (2012) document a deployed Stackelberg model of randomized Coast Guard port patrols. Washburn and Wood (1995) model network interdiction as a two-person zero-sum game. Schelling (1966) represents escalation as a shared, manipulated risk, motivating our incident hazard.

What we add: we combine maritime allocation, evolving control, and shared escalation risk in a general-sum game where both sides contest control simultaneously and war costs both. Computational rules support comparisons of strategies and designs across thousands of runs. Numerical parameters are our assumptions (Table 1), not values derived from documented literature.

3 Game formulation

One period t (timeline): (1) Nature draws the sea state w_r^t of each region (scenario ω). (2) Each player observes its own ships, ℓ^t, c^t, w^t, and detects opponent ships w.p. $q_{i,v}$. (3) Ships whose endurance is nearly spent are forced toward homeor toward their side's oiler. Ships due for maintenance or with an engineering casualty are forced home. (4) Both players choose moves and postures

simultaneously. (5) Moves resolve under movement budgets, and endurance updates. Maintenance clocks, engineering casualties and repairs update, and each oiler moves. (6) Control indices update, incidents are drawn, and ℓ moves up or down. (7) Stage payoffs are paid, and the game stops if $\ell = L$ (war) or $t = T = \sout{60}180$.

Symbol	Meaning	Value
T; δ	periods (2 days); discount	$\sout{60}180$; 0.98
N_k	fleet size	10 DDG; 8 PLAN, 6 CCG
D_k	endurance (periods)	10 DDG; 8 PLAN; 6 CCG
F_i	movement budget (moves/period)	8 Blue; 10 Red
w_k	strength per hull (rough seas)	1 DDG, PLAN; 0.6 CCG (0.3)
α	assert multiplier	1.5
κ; λ	presence damping; control rate	1; 0.3
h_0	baseline incident hazard	0.006
$h^{\mathsf{B}}_{\mathsf{A}}$; $h^{\mathsf{R}}_{\mathsf{A}}$	assert hazard	0.03; 0.036/0.012 PLAN/CCG

Symbol	Meaning	Value
L; ℓ^1	war level; starting level	5; 1
μ	chance quiet period lowers ℓ	0.25 (D3: 0.5 at $\ell \geq 3$)
γ; η	move; tension cost	0.1; 1
W	war penalty (both sides)	200
$q_{i,v}$	detection: here / nearby / far	0.95 / 0.6 / 0.2, $\times$0.7 rough
Λ; β	MIP: incident price; persistence	20; 0.96
ε_i	escalation-risk budget	0.05 / 0.15 / 0.40
ζ_D	redesign cost per period	D1 0.3; D2 0.5; D3 0
M_{on}; M_{in}	periods operating; in maintenance	90; 45
m_0	share in maintenance at $t = 1$	1/3
p_c; Z_c	casualty chance per period at sea; repair periods	$1 - 0.99^2 \approx 0.02$; 5
D_o	oiler endurance (periods)	15

Table 1: Parameter values, all of them our assumptions (no public data). Hazards and μ were set so that war is uncommon but not rare (0–34% of games).

3.1 Players, map and time

Players: $\mathcal{I} = \{\mathsf{B}, \mathsf{R}\}$. Blue is the U.S. Navy, $\mathcal{K}_{\mathsf{B}} = \{\text{DDG}\}$. Red is China, with navy ships and coast guard or maritime militia hulls, $\mathcal{K}_{\mathsf{R}} = \{\text{PLAN}, \text{CCG}\}$ (fleets in Table 1).

Map: the theater is an undirected graph $G = (\mathcal{V}, \mathcal{E})$. Its nodes are home ports, where ships refuel, and eight contested zones $\mathcal{C} = \{\text{SEN}, \text{MIY}, \text{TWN}, \text{EOT}, \text{LUZ}, \text{SCB}, \text{STS}, \text{PAR}\}$ (Senkakus, Miyako Strait, Taiwan Strait, east of Taiwan, Luzon Strait, Scarborough Shoal, Second Thomas Shoal, Paracels). Blue's home ports are $H_{\mathsf{B}} = \{\text{YOK}, \text{GUA}\}$ (Yokosuka, Guam) and Red's are $H_{\mathsf{R}} = \{\text{NBO}, \text{SAN}\}$ (Ningbo, Sanya)[1]. An edge joins two nodes a ship can sail between in one period. The zones form three regions that share weather: North $= \{\text{SEN}, \text{MIY}\}$, Center $= \{\text{TWN}, \text{EOT}, \text{LUZ}\}$, South $= \{\text{SCB}, \text{STS}, \text{PAR}\}$. Zone v is worth V_v^i to player i (Table 2, e.g. Scarborough Shoal and the Paracels matter more to Red).

Time: periods $t = 1, \ldots, T$ of two days, $T = \sout{60 \text{ (about four months)}}180$ (one year), discounted by δ per period. All ships start in port with full endurance, split evenly between each side's two ports. A third of each fleet starts in maintenance (§3.4).

Zone	TWN	SEN	MIY	EOT	LUZ	SCB	STS	PAR
V_v^{B}	10	6	4	4	6	5	5	2
V_v^{R}	10	6	3	4	5	7	6	6
c_v^1	0	0.2	0.3	0.2	0.1	-0.2	0	-0.4

ω	calm	nominal	typhoon	extreme
π_ω	0.25	0.40	0.25	0.10
$p_s(\omega)$	0	0.02	0.06	0.12
$p_r(\omega)$	0	0.20	0.35	0.50

Table 2: Our assumed values. Left: zone values and starting control. Right: scenario weight π_ω and per-period chances of a storm starting $p_s(\omega)$ and of rough seas $p_r(\omega)$.

3.2 State, information and weather

State: $s^t = (x^t, d^t, c^t, \ell^t, w^t, r^t, z^t, e^t, o^t, d^{o,t})$. Here $x^t_{k,v} \in \mathbb{Z}_{\geq 0}$ is the number of type-k ships at node v. The *endurance* $d^t_j \leq D_k$ is the number of periods ship j can still stay at sea (down one per period at sea, refilled after a full period in port). The *control index* $c^t_v \in [-1, 1]$ of each zone is $+1$ when Blue is fully in charge and -1 when Red is. The *escalation level* $\ell^t \in \{0, \ldots, L\}$ runs from calm to armed conflict. The sea state of region r is $w^t_r \in \{\text{calm}, \text{rough}, \text{storm}\}$. Each ship j also carries a readiness state: r^t_j is the number of operating periods left before it is due for maintenance, z^t_j is the number of periods it still has to spend in port (maintenance or repair, 0 when it is available), and $e^t_j \in \{0, 1\}$ is 1 if it has an engineering casualty and is heading home. Each player's oiler is at node o^t_i with $d^{o,t}_i$ periods of its own fuel left (§3.4).

Information: each player sees its own ships and the public c^t, ℓ^t, w^t, but not where the opponent's ships are now. Each period it detects the opponent at node v with probability $q_{i,v}$ (Table 1, for an own ship there, a ship next door or v near a home port, or satellite only), seeing the count of each type and the posture. Its *belief* $\hat{x}^t_{-i,v}$ is the last count it detected at v (exact at the start, when all ships are in port), so moves are chosen at information sets that do not reveal where the opponent is now.

Weather: a random factor neither side controls. In each of four scenarios (not calendar seasons) $\omega \in \Omega = \{\text{calm, nominal, typhoon, extreme}\}$ with weights π_ω, each period a region not already in a storm enters one with probability $p_s(\omega)$, and is otherwise rough with probability $p_r(\omega)$ (Table 2). A storm lasts three periods. It closes the region's zones (no entry, ships inside stay), stops incidents there and freezes their control. Rough seas halve CCG strength and degrade detection.

3.3 Actions and constraints

Actions: each period both players choose, simultaneously, a destination for each ship (stay, or sail one edge) and a posture $p_{i,v} \in \{\text{P}, \text{A}\}$ in each zone v where they will have ships. *Patrol* is routine presence. *Assert* stands for the §1 actions, treated as one posture. Asserting multiplies the side's strength there by α but makes incidents more likely (§3.5).

[1] NBO and SAN stand for the Ningbo–Zhoushan and Hainan (Yulin) basing areas. CCG and militia ports are folded into them.

Constraints: with $f^t_{k,uv}$ the number of type-k ships sailing $u \to v$ in period t and $\mathcal{N}(u)$ the neighbors of u,

$$x^{t+1}_{k,u} = x^t_{k,u} + \sum_{v\in\mathcal{N}(u)} f^t_{k,vu} - \sum_{v\in\mathcal{N}(u)} f^t_{k,uv} \qquad \text{(flow conservation: no ships are added or removed)}$$

$$\sum_{k\in\mathcal{K}_i} \sum_{(u,v)} f^t_{k,uv} \le F_i \qquad \text{(movement budget: at most } F_i \text{ moves per period, Table 1)}$$

where the theater-wide movement budget F_i stands in for operational requirements and station-keeping. *Endurance:* a ship may not sail to a node from which it could not ~~get home~~reach a refuel point on its remaining endurance, unless the move brings it closer to ~~home~~one. The refuel points of player i are its home ports and its oiler when the oiler is in a zone, $\mathcal{R}^t_i = H_i \cup \{o^t_i : o^t_i \in \mathcal{C}\}$. A ship uses whichever is closer, at distance $\mathrm{dist}(v,\mathcal{R}^t_i) = \min_{u\in\mathcal{R}^t_i} \mathrm{dist}_i(v,u)$. A ship whose endurance has fallen to its distance from ~~home~~the nearest refuel point is on *forced return*. It sails one edge toward ~~port~~that point each period, then spends one full period in port or alongside the oiler to refit, after which d_j resets to D_k. Forced moves use the movement budget first, and the player decides only for its *free* ships. *Ports, storms and infeasible orders:* no ship may enter opponent home ports or storm regions, and orders over budget are dropped.

3.4 Readiness: maintenance, engineering casualties and oilers

A year-long game needs the things that keep a real fleet from being at sea all the time.

Maintenance: a ship operates $M_{\mathrm{on}} = 90$ periods (six months) and then spends $M_{\mathrm{in}} = 45$ periods (three months) in maintenance at a home port, so in steady state a third of each fleet is unavailable. While a ship is available its maintenance clock runs down, $r^{t+1}_j = r^t_j - 1$. When $r^t_j \le 0$ the ship is on forced return to its nearest home port (not to the oiler). On arrival it enters maintenance, $z_j = M_{\mathrm{in}}$, and z_j falls by one each period. When z_j reaches 0 the ship is available again with $r_j = M_{\mathrm{on}}$ and full endurance $d_j = D_k$. A ship with $z_j > 0$ stays in port and is not part of either player's decision.

Staggered start: at $t = 1$ a share $m_0 = 1/3$ of each player's ships, chosen at random, is in maintenance with z^1_j drawn uniformly from $\{1,\dots,M_{\mathrm{in}}\}$. Every other ship has r^1_j drawn uniformly from $\{1,\dots,M_{\mathrm{on}}\}$, so ships come due at different times and not all at once. These draws use their own random stream, so a given seed has the same maintenance schedule for every pair of policies.

Engineering casualties: each day at sea a ship suffers an engineering casualty with probability 1%, so in a two-day period the chance is $p_c = 1 - 0.99^2 \approx 0.02$. Then $e_j = 1$ and the ship is on forced return to its nearest home port. On arrival it is repaired for $Z_c = 5$ periods (ten days), $z_j = Z_c$, after which $e_j = 0$ and $d_j = D_k$. A ship that has a casualty while heading in for maintenance is repaired during the maintenance period.

Oilers: each player has one oiler. It is not commanded ship by ship. It follows a fixed rule that keeps it where its side is doing well. The side's *best region* is the one with the highest value-weighted control from that side's point of view,

$$r^{*,t}_i = \arg\max_{r\in\{\mathrm{N,C,S}\}} \sigma_i \frac{\sum_{v\in r} V^i_v c^t_v}{\sum_{v\in r} V^i_v}.$$

The oiler picks a zone of $r^{*,t}_i$ uniformly at random as its target, sails one edge per period toward it, and picks a new random target when it arrives or when the best region changes. So it wanders among the zones of the best region and follows when that region changes. Its own fuel d^o_i falls by one per period away from port. When $d^{o,t}_i \le \mathrm{dist}_i(o^t_i, H_i)$ it heads for its nearest home port, and one period there refills it to $D_o = 15$ (30 days). A ship that spends a full period in the oiler's zone without moving is refueled to D_k, as in port. The oiler obeys the same storm and port rules as warships. It adds no strength and cannot be attacked yet.

3.5 How the state changes: control, incidents and escalation

After the moves, steps 1–3 apply in each contested zone v not in a storm. Step 4 updates ℓ once for the theater.

1. Local strength: each side's weight of presence in the zone is $S_{i,v} = \alpha^{\mathbf{1}[p_{i,v}=\mathrm{A}]} \sum_{k\in\mathcal{K}_i} w_k x_{k,v}$, its hulls weighted by w_k (a CCG hull counts less than a navy ship, less still in rough seas), times α when asserting.

2. Control: the *local balance* $\phi_v = (S_{\mathrm{B},v} - S_{\mathrm{R},v})/(S_{\mathrm{B},v} + S_{\mathrm{R},v} + \kappa)$ lies between -1 and 1 and is positive when Blue is locally stronger. The constant $\kappa = 1$ keeps a small presence from counting as full control (e.g. one DDG alone gives $\phi_v = 0.5$). Control is earned by staying present and fades when a side leaves. Each period it moves a fraction $\lambda = 0.3$ of the way toward the local balance,

$$c^{t+1}_v = (1-\lambda)\, c^t_v + \lambda\, \phi_v. \tag{1}$$

3. Incidents: an incident (a collision, an unsafe interaction, negative media coverage) can happen only where both sides have ships. It occurs with probability $\rho_v = 1 - e^{-h_v}$, where the *hazard* h_v is

$$h_v = \left(h_0 + h^{\mathrm{B}}_{\mathrm{A}}\, \mathbf{1}[p_{\mathrm{B},v} = \mathrm{A}] + h^{\mathrm{R}}_{\mathrm{A},v}\, \mathbf{1}[p_{\mathrm{R},v} = \mathrm{A}]\right) \cdot \min\{S_{\mathrm{B},v}, S_{\mathrm{R},v}\} \cdot \left(1 + \ell^t/2\right). \tag{2}$$

(i) A small baseline hazard h_0, plus a term for each side that asserts. Red's assert hazard $h^{\mathrm{R}}_{\mathrm{A},v} = \theta_v h^{\mathrm{R}}_{\mathrm{A,PLAN}} + (1-\theta_v) h^{\mathrm{R}}_{\mathrm{A,CCG}}$ blends its navy and coast guard values by θ_v, the PLAN share of Red's strength at v, so coast guard pressure is less escalatory. (ii) More ships in contact mean more encounters, limited by the smaller force. (iii) Incidents are likelier when tension is high ($2.5\times$ at $\ell = 3$).

4. Escalation: each incident raises ℓ by one. A period with no incident anywhere lowers ℓ by one with probability μ (Table 1). If ℓ reaches L, the crisis has become an armed conflict and the game ends.

3.6 Payoffs

Stage payoff: at the end of period t, player i receives

$$g_i^t = \sigma_i \sum_{v \in \mathcal{C}} V_v^i \, c_v^{t+1} - \gamma m_i^t - \eta \, \ell^{t+1} - \zeta_D \mathbf{1}[i = \mathsf{B}]. \tag{3}$$

Control: $\sigma_{\mathsf{B}} = +1$ and $\sigma_{\mathsf{R}} = -1$ turn the shared control index into each side's gain (e.g. Blue's control term is $V_v^{\mathsf{B}} c_v$, positive where Blue holds the zone and negative where Red does). *Fuel:* m_i^t ship moves in period t, at γ each. *Tension:* η per escalation level per period, for diplomatic and economic strain. *Redesign cost:* ζ_D is Blue's redesign cost per period (§3.10).

Total payoff: the game ends at τ, the first period with $\ell^\tau = L$ (war), or at T. Player i's payoff is the discounted sum of its stage payoffs, minus a war penalty W (Table 1) if the game ends in war:

$$u_i = \sum_{t=1}^{\tau} \delta^{t-1} g_i^t - W \, \mathbf{1}[\ell^\tau = L]. \tag{4}$$

Both sides pay W, so the game is general-sum, not zero-sum. Both prefer to avoid war. For scale, Blue earns at most $\sum_v V_v^{\mathsf{B}} = 42$ in a period in which it fully controls every zone, so a war ($W = 200$) costs roughly five such periods.

3.7 How a player decides: the best-response MIP with a risk budget

Our main decision rule, *MIP-BR*, is a best response to the opponent's believed positions that caps incident risk, written as a mixed-integer program and solved with Gurobi each period. It takes as given its beliefs $\hat{x}_{-i}$, the sea state, ℓ^t, its own ships, and, in each zone, the opponent posture it saw on its last detection there (patrol if never detected). Forced-return ships are fixed ($\bar{x}_{k,v}$, by destination).

Decision variables: $y_{j,w} \in \{0,1\}$ is 1 if free ship j goes to node w (only moves allowed by §3.3). Control and incident risk are nonlinear in ship counts, so we list every count a zone could end up with. Option $b_{v,(n,p)} \in \{0,1\}$ is 1 if the player will have $n = (n_k)_{k \in \mathcal{K}_i}$ hulls in zone v with posture p. Each option's value and risk are computed in advance, which keeps the program linear.

Value of an option:

$$a_v(n,p) = V_v^i \beta \, \sigma_i \phi_v(n,p) - \Lambda \rho_v(n,p), \qquad \beta = \frac{\lambda}{1 - \delta(1-\lambda)} \approx 0.96.$$

with ϕ_v, ρ_v (§3.5) at believed opponent positions. Here β is the discounted effect on control of one period of balance, and Λ is the incident price.

The program:

$$\begin{aligned}
\max_{y,b} \quad & \textstyle\sum_{v \in \mathcal{C}} \sum_{(n,p)} a_v(n,p) \, b_{v,(n,p)} - \gamma \sum_j \sum_{w \neq \mathrm{loc}(j)} y_{j,w} - 1000\,\xi && \text{(control value} - \text{incident cost} - \text{fuel cost)} \\
\text{s.t.} \quad & \textstyle\sum_w y_{j,w} = 1 \quad \forall j && \text{(each free ship gets one destination)} \\
& \textstyle\sum_j \sum_{w \neq \mathrm{loc}(j)} y_{j,w} \le F_i - \text{(forced moves)} && \text{(movement budget)} \\
& \textstyle\sum_{(n,p)} b_{v,(n,p)} = 1 \quad \forall v && \text{(one option per zone)} \\
& \textstyle\sum_{(n,p)} n_k \, b_{v,(n,p)} = \bar{x}_{k,v} + \sum_{j \text{ of type } k} y_{j,v} \quad \forall v,k && \text{(the option matches the moves)} \\
& \textstyle\sum_{v \in \mathcal{C}} \sum_{(n,p)} \rho_v(n,p) \, b_{v,(n,p)} \le \varepsilon_i (1 - \ell^t / L) + \xi && \text{(escalation-risk budget)} \\
& y, b \text{ binary}, \ \xi \ge 0.
\end{aligned}$$

The escalation-risk budget: the left side of the last constraint is the expected number of incidents this period. The cap ε_i shrinks to zero as ℓ approaches war and defines the feasible region, the allocations the player considers safe enough. The proposal's conservative, moderate and aggressive commanders become three budgets (Table 1), and RQ1 varies ε_i. A ship on forced return may have to pass through a zone the opponent occupies. That risk cannot be avoided, and near war it can exceed the cap. The slack $\xi \ge 0$, penalized at 1000 per unit, lets the program exceed the cap by only that amount, so it always has a solution.

3.8 The best-response MIP for the larger fleets (proposed)

The program above is exact, but it does not scale to the Workshop 2 fleets. This subsection shows why and gives the version we propose. The rules of the game (§3.3–§3.6) do not change. Only the way the player's decision problem is written down changes.

Why the Workshop 1 program breaks: it lists every count vector a zone could end up with. If $A_{k,v}$ ships of type k could arrive at zone v this period, the number of options in that zone is

$$|\{b_{v,(n,p)}\}| \;=\; 2 \prod_{k \in \mathcal{K}_i} \big(A_{k,v} + 1\big).$$

In Workshop 1, $|\mathcal{K}_{\mathsf{B}}| = 1$ and $|\mathcal{K}_{\mathsf{R}}| = 2$, so this was a few hundred. With the eight Blue types and, for example, $A_{k,v} = (12, 13, 16, 12, 2, 4, 9, 6)$ (US destroyers, Japanese destroyers, Japanese frigates, Japan Coast Guard, LCS, Philippine frigates, Philippine patrol ships, Philippine Coast Guard), it is $2 \cdot 13 \cdot 14 \cdot 17 \cdot 13 \cdot 3 \cdot 5 \cdot 10 \cdot 7 \approx 8 \times 10^7$ options in one zone. The model cannot be built, whatever the license or the run time.

Step 1, moves as flows (exact): free ships of the same type, at the same node, with the same set of allowed destinations are interchangeable. Call such a set a group g, with type $k(g)$, location $u(g)$, allowed destinations $\mathcal{W}_g$ and size $|g|$. Replace the per-ship binaries $y_{j,w}$ by integer flows

$$f_{g,w} \in \mathbb{Z}_{\geq 0} \quad (w \in \mathcal{W}_g), \qquad \sum_{w \in \mathcal{W}_g} f_{g,w} = |g|.$$

These are the flows $f^t_{k,uv}$ of §3.3. Every assignment of ships gives a flow and every flow gives an assignment, so this step changes nothing.

Step 2, options by strength level (exact for control): control depends on the counts only through the side's strength. With weights $w_k(w_r)$ for the sea state (small hulls halved in rough seas), the own un-asserted strength in zone v is

$$R_v = \sum_{k \in \mathcal{K}_i} w_k \left(\bar{x}_{k,v} + \sum_{g:k(g)=k} f_{g,v} \right).$$

Every weight, halved or not, is a multiple of 0.05, so $R_v = 0.05\,m$ for an integer m. The option $b_{v,(m,p)} \in \{0,1\}$ means "strength level m and posture p in zone v", and the matching constraint becomes

$$\sum_{(m,p)} b_{v,(m,p)} = 1, \qquad \sum_{(m,p)} 0.05\,m\, b_{v,(m,p)} = R_v.$$

The option values are the same formulas as before, $a_v(m,p) = V_v^i \beta\, \sigma_i \phi_v(m,p) - \Lambda \rho_v(m,p)$, with own strength $S_{i,v} = \alpha^{\mathbf{1}[p=\mathrm{A}]}\, 0.05\,m$. A zone now has at most $2(M_v+1)$ options, where $0.05\,M_v$ is the largest strength that could gather there (a few hundred options, not 10^7). Because ϕ_v depends on the counts only through $S_{i,v}$, control values are exactly those of the Workshop 1 program.

The one approximation (incident risk when asserting with a mixed force): the assert hazard blends the navy and coast guard values by the navy share of the side's strength in the zone, $h^i_{\mathrm{A},v} = \theta_{i,v} h^i_{\mathrm{A,navy}} + (1-\theta_{i,v}) h^i_{\mathrm{A,cg}}$ (§3.5). A strength level does not fix $\theta_{i,v}$. Inside the program the player uses $\hat{\theta}_{i,v}$, the navy share of its own ships already in zone v or fixed to arrive there (its whole available fleet if there are none), so $\rho_v(m,p)$ is computed with $\hat{\theta}_{i,v}$. This affects only the player's own estimate of incident risk, and only in zones where it asserts with both navy and coast guard ships. The game engine still draws every incident with the true mix.

The program:

$$\begin{aligned}
\max_{f,b} \quad & \textstyle\sum_{v \in \mathcal{C}} \sum_{(m,p)} a_v(m,p)\, b_{v,(m,p)} - \gamma \sum_g \sum_{w \neq u(g)} f_{g,w} - 1000\,\xi && \text{(control value} - \text{incident cost} - \text{fuel cost)} \\
\text{s.t.} \quad & \textstyle\sum_{w \in \mathcal{W}_g} f_{g,w} = |g| \quad \forall g && \text{(every free ship gets one destination)} \\
& \textstyle\sum_g \sum_{w \neq u(g)} f_{g,w} \leq F_i - (\text{forced moves}) && \text{(movement budget)} \\
& \textstyle\sum_{(m,p)} b_{v,(m,p)} = 1 \quad \forall v && \text{(one option per zone)} \\
& \textstyle\sum_{(m,p)} 0.05\,m\, b_{v,(m,p)} = \sum_k w_k \left(\bar{x}_{k,v} + \sum_{g:k(g)=k} f_{g,v} \right) \quad \forall v && \text{(the option matches the moves)} \\
& \textstyle\sum_{v \in \mathcal{C}} \sum_{(m,p)} \rho_v(m,p)\, b_{v,(m,p)} \leq \varepsilon_i \left(1 - \ell^t / L\right) + \xi && \text{(escalation-risk budget)} \\
& f \text{ integer},\ b \text{ binary},\ \xi \geq 0.
\end{aligned}$$

Compared with §3.7, y becomes f, and the index n (a count for each type) becomes m (a strength level). Everything else is unchanged.

Check before we switch: on the Workshop 1 fleet, where both programs can be built, we solve both every period on the same seeds and report (i) how often they choose a different plan, (ii) the gap in the objective, and (iii) the payoffs and war rates of MIP-BR against MIP-BR. Blue has one type there, so any difference comes from Red's mix of PLAN and CCG, which is exactly the approximation above.

3.9 The template game and the policies we compare

Template game: the full game has far too many strategies to solve for an equilibrium directly, so we also study a small strategic-form version. Each player picks one of 14 *templates*, which pair one of seven ways to split ~~70%~~two thirds (the share out of maintenance) of its fleet across North/Center/South (60/20/20, 40/40/20 or 34/33/33 percent and their permutations) with one posture (P or A) in all its zones. Ships are spread evenly within a region, Red's PLAN and CCG in proportion, and ~~fractional hulls are allowed~~only whole ships are used. With N_k^{dep} the deployable ships of type k rounded to a whole number and π_r the template's share for region r, the region counts are the largest-remainder rounding of $N_k^{\mathrm{dep}} \pi_r$. Each region gets $\lfloor N_k^{\mathrm{dep}} \pi_r \rfloor$ ships, and the ships left over go one at a time to the regions with the largest fractional parts. The same rule splits a region's ships over its zones. If both kept their templates, control would settle at $c_v = \phi_v$ (set $c_v^{t+1} = c_v^t$ in (1)), so a template pair's payoff is that settled control minus the incident cost, in calm seas at $\ell = 1$: $M_i(a_{\mathrm{B}}, a_{\mathrm{R}}) = \sigma_i \sum_{v \in \mathcal{C}} V_v^i \phi_v - \Lambda \sum_{v \in \mathcal{C}} \rho_v$. This gives a finite game $\Gamma = \langle \mathcal{I}, (S_i), (M_i) \rangle$ with $|S_i| = 14$, which we analyze with iterated elimination of strictly dominated strategies (IESDS), pure-NE enumeration, and the maximin and correlated-equilibrium LPs.

Policies (§4): *Random* picks moves and postures at random, as a benchmark. *Rule-based* (moderate commander) sends ships where value per unit of opponent strength is highest, asserts only where it believes it outnumbers the opponent and $\ell \leq 2$, then pulls back until its expected incidents fit $\varepsilon = 0.15\,(1 - \ell/L)$. *Greedy* is a fast heuristic for the MIP (same objective and budget, no solver). *MIP-BR*(ε) solves the MIP exactly. *Equilibrium* plays its part of the pure NE of Γ with the higher total payoff (our choice), steering

free ships toward the template's split (more ships to a region's more valuable zones). *Maximin* draws a template every three periods from its maximin mixed strategy of Γ. The two template policies switch to patrol when $\ell > 3$ and drop assertions over their risk budget.

3.10 Resilient redesigns (Blue)

A redesign changes Blue's system. D1 is structural (an alternative path), D2 and D3 operational (monitoring, rapid response). Each is compared with the baseline D0 in every scenario. **D1 dispersed basing:** a third Blue port at Subic Bay, linked to SCB, LUZ and STS, giving alternative paths and shorter trips South. Blue starts 4/3/3 at YOK/GUA/Subic. **D2 redundant sensing:** Blue detects in every zone with probability at least 0.6, even in bad weather. **D3 de-escalation measures:** once $\ell \geq 3$, a quiet period lowers ℓ with probability 0.5 instead of 0.25, and assert hazards are halved, which helps Red as much as Blue. Costs ζ_D are in Table 1.

Simplifying assumptions: there are two unitary players, control is a force-ratio index, incidents are the only path to war, beliefs are last-seen counts, and best responses look one period ahead. Oilers follow a fixed rule and cannot be targeted, maintenance takes place only at home ports, and engineering casualties are independent from ship to ship.

4 Preliminary results

Note: the numbers in this section are still the Workshop 1 results (60 periods, fractional hulls, no readiness). They will be replaced by the rerun with the changes above.

§4.1 is analytical. The rest use 200 seeds per configuration. Unless stated, Red plays MIP-BR with $\varepsilon_{\mathsf{R}} = 0.15$ in nominal weather.

4.1 Template game and a small example: patrol or assert, and sustainable restraint

IESDS reduces Γ from 14×14 to 2×4. Blue keeps only the center-heavy split (20/60/20), and Red keeps the south-heavy and center/south-heavy splits. Of the two pure NE, the *equilibrium* policy plays (Blue center-heavy assert, Red south-heavy patrol), $(0.39, -9.27)$, over (Blue patrol, Red center/south assert), $(-14.44, 2.71)$. Without the escalation cost, the zero-sum maximin LP has a pure equilibrium where *both* sides concentrate on the Center and assert. The shared incident cost is what makes at most one side assert. Fixing Blue center-heavy and Red south-heavy, each side only chooses P or A. The resulting 2×2 sub-matrix of Γ, normalized so (P,P) $= (0,0)$, has (P,A) $= (-9.61, 1.20)$, (A,P) $= (0.28, -10.45)$ and (A,A) $= (-11.41, -11.33)$ (Blue, Red). Asserting against a patroller pays a little (one-period gain $G_{\mathsf{B}} = 0.28$, $G_{\mathsf{R}} = 1.20$), but mutual assertion costs more than patrolling alone, so $BR_i(\mathrm{P}) = \mathrm{A}$ and $BR_i(\mathrm{A}) = \mathrm{P}$. There are two pure NE, (A,P) and (P,A), and a mixed NE (Blue asserts w.p. 0.58, Red 0.13) with mutual assertion w.p. 0.08 and payoffs $v_{\mathsf{B}} = -1.30$, $v_{\mathsf{R}} = -6.02$. The welfare-maximizing correlated equilibrium (CE LP) puts *zero* on (A,A). A mediator (e.g. a hotline) removes mutual assertion.

In the repeated game, suppose both patrol, and any assertion makes both fall back to the mixed NE for good. By the one-stage deviation principle, patrolling is optimal iff $(1-\delta)G_i + \delta v_i \leq 0$, i.e. $\delta \geq G_i/(G_i - v_i)$, which is 0.18 for Blue and 0.17 for Red. With $\delta = 0.98$, **mutual restraint is a subgame perfect equilibrium**, though our one-period-ahead policies cannot see this.

4.2 RQ1: control versus war risk, and the risk-budget game

B\R	cons. 0.05	mod. 0.15	aggr. 0.40
cons. 0.05	−194, 199	−303, 315	−308, 318
	war 0.10	0.11	0.12
mod. 0.15	18, −36	−144, 101	−192, 140
	war 0.15	0.20	0.21
aggr. 0.40	74, −80	−83, 16	**−151, 39**
	war 0.14	0.25	**0.34**

Table 3: The risk-budget game: $(u_{\mathsf{B}}, u_{\mathsf{R}})$ and war rate when both sides choose ε (s.e. $\approx$ 5–12). Bold: the unique NE.

Frontier: Raising ε_{B} from 0.02 to 0.15 lifts Blue's control share from -0.25 to -0.05 while the war rate rises from 0.07 to 0.20 (Red fixed at 0.15). Going to 0.40 adds only 0.06 of control. Most of what a looser budget buys is bought by $\varepsilon \approx 0.15$–0.25.

Risk-budget game (Table 3)**:** each side picks a budget. Whatever the other side picks, "aggressive" pays each side the most, so both pick it, and (aggressive, aggressive) is the unique pure NE. It also has the *highest* war rate (0.34), and both sides would do at least as well if both chose moderate ($-144, 101$ vs. $-151, 39$, war 0.20). This is a prisoner's dilemma, so restraint must come from repetition or coordination (§4.1).

4.3 RQ2: strategic versus heuristic play

Blue \ Red	rule-based (moderate)	MIP-BR	equilibrium
random	−283 / −0.14 / 0.15	−319 / −0.16 / 0.33	−164 / −0.08 / 0.06
rule-based (moderate)	−122 / −0.05 / 0.04	−184 / −0.09 / 0.06	35 / +0.06 / 0.04
greedy	−70 / +0.01 / 0.12	−137 / −0.05 / 0.14	65 / +0.08 / 0.03
MIP-BR	−75 / +0.01 / 0.13	−144 / −0.05 / 0.20	53 / +0.08 / 0.08
equilibrium	−72 / −0.01 / 0.07	**−116** / −0.03 / 0.10	21 / +0.05 / 0.06
maximin	−170 / −0.09 / 0.04	−221 / −0.12 / 0.08	−35 / −0.01 / 0.00

Table 4: Blue payoff u_{B} / control share / war rate (s.e. of u_{B} is 1–7). Nominal weather, design D0. The MIP-BR column is each policy's worst case.

Greedy, MIP-BR and equilibrium beat the random and rule-based benchmarks (Table 4). The **equilibrium** policy has the best worst case (-116) and, against MIP-BR, half the war rate of MIP-BR itself (0.10 vs. 0.20). Greedy and MIP-BR do best against weaker opponents but escalate more. **Maximin** is the safest (war rate ≤ 0.08) but gives up the most control, guarding against templates Red rarely plays. Greedy matched the MIP optimum in 67% of 1,524 Blue decisions but, unlike the MIP, broke the risk budget in 14%.

4.4 RQ3: robustness to weather and resilient designs

Design	calm	nom.	typh.	extr.	exp.	worst
D0 baseline	−164/.24	−144/.20	−134/.12	−119/.10	−144	−164
D1 dispersed	−153/.33	−141/.28	−131/.24	**−98**/.14	−137	−153
D2 sensing	−181/.23	−164/.23	−149/.12	−136/.11	−162	−181
D3 de-escalation	**−140**/.10	**−118**/.05	**−111**/.02	−102/.01	**−120**	**−140**

Table 5: Blue designs by weather scenario (MIP-BR vs. MIP-BR): u_B / war rate (s.e. ≈ 7).

Weather: bad weather helps Blue slightly. Most policies do worst in *calm* seas, since rough seas halve Red's coast-guard strength and storms pause operations and incidents (MIP-BR war rate 0.24 calm, 0.10 extreme). The policy ranking is robust. MIP-BR ($\varepsilon = 0.40$) and equilibrium are first and second in all four scenarios, so the best policy on average is also best in the worst case. The top scorer also has the most wars (0.12–0.29), so $W = 200$ may be too small.

We score each policy or design x by its expected payoff $\sum_\omega \pi_\omega J(x,\omega)$ and worst case $\min_\omega J(x,\omega)$.

Designs: De-escalation measures (D3) are best in expected and worst-case terms and cut the war rate fourfold (0.20 → 0.05, nominal). They also raise Red's payoff (101 → 144), since fewer wars help both sides. We charged them nothing, so part of this win is by assumption. Workshop 2 will give them a cost. Dispersed basing (D1) barely changes Blue's payoff but cuts Red's (101 → 16), at a higher war rate (0.28). Redundant sensing (D2) shows no measurable benefit even at zero cost, because better Blue information barely changes Blue's decisions in this model. **Limitations:** parameters are assumptions, and policies look one period ahead. The war rate depends strongly on h, μ and W, and payoff differences below about 20 are ties (s.e. 5–12).

5 Prototype

Current state: `wargame_gui.py` runs the model of §3 as a **GUI** (pick each side's plan, Blue's design and the weather, then step through turns or run many games), as **batch** runs, or as a **trace**. `experiments.py all` regenerates every number in §4.

How the code reflects the model: `Game.step` is the timeline of §3 with (1)–(4), `MIPBRPolicy` solves the MIP of §3.7 each period, and `MetaGame` builds Γ and runs IESDS, NE enumeration and both LPs. **Still to build:** learning agents, an attacker that reacts to Blue, Bayesian beliefs and multi-period policies (§6).

6 Plan for Workshop 2

Model: Bayesian beliefs, multi-period best response, sensitivity analysis on V, h, W, μ, and calendar-based weather from real data. Reformulate the movement budget F_i with considerations for oilers. **Learning:** a Red agent that learns from repeated play, and other learning methods on Γ. Do they converge, drift toward restraint, and exploit rule-based Blue? **Attacks and defenses:** three Red attacks are coded. A coast guard blockade at SCB and STS blocks Blue transits, jamming halves Blue's detection in one region, and false contacts appear in Blue's picture. We will add a Red attacker that reacts to Blue's posture and test randomized patrols, D1 and D2 as defenses. **Designs:** D0–D3 and combinations under attack and a long typhoon, and a cost for D3.

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